

SHANTELE WAMBUI KUNGU

Data Analyst | AI/ML Engineer | Operational Intelligence

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PROFESSIONAL SUMMARY

Data Analyst and AI/ML Engineer with production experience building end-to-end intelligent systems across fraud detection, predictive maintenance, safety monitoring, and operational analytics. Co-built ClaimGuard - a production-grade insurance fraud detection platform achieving AUC-ROC 0.84, cutting claim review time from 5-15 days to under 30 seconds, and projecting KES 148M in annual savings for a mid-size insurer. Full-stack Software Engineering Intern at E&M Techhouse (INUKA/PLP), shipping features across six live products.

TECHNICAL SKILLS

ML & Data Science: XGBoost, scikit-learn, SHAP, pandas, NumPy, SMOTE, Plotly, Matplotlib

Data Engineering: ETL pipeline design, PostgreSQL, SQLAlchemy, Alembic, Great Expectations

Languages: Python, SQL, TypeScript, JavaScript (ES6+), Dart

Frameworks: FastAPI, Streamlit, Flask, Spring Boot, React, Next.js 14, Flutter

Infrastructure: Docker, Docker Compose, Celery, Redis, GitHub Actions, Render, Vercel

AI/LLM: Groq LLM integration, SHAP explainability, XGBoost model registry

PROFESSIONAL EXPERIENCE

Software Engineering Intern - E&M Techhouse, Nairobi (INUKA/PLP) | 2025 - Present

ClaimGuard - AI Insurance Fraud Detection (FastAPI + Next.js)

- * Co-built production fraud detection platform: AUC-ROC 0.84, Precision 79%, Recall 81%
- * Engineered 25 features; network/ring detection found 12 fraud rings (KES 4.2M exposure)
- * Reduced processing time from 5-15 days to under 30 seconds - 95% efficiency gain
- * Built FastAPI backend: ingestion, ETL, rules engine, ML scoring, audit trail
- * Built Next.js frontend: case review, HITL queue, network visualisation, ML admin

Sentinel - HSE Early-Warning Platform (Next.js + Spring Boot)

- * Built AI HSE Reporting Agent: Groq LLM integration, 15-section regulatory PDF generator
- * Built context-aware AI chatbot for Q&A over live HSE event data

Ecosystem Banking Platform - Mobile, Frontend & Backend

- * Sole Flutter developer: biometric auth, offline-first sync, follow-up calendar, push notifications
- * React/Next.js: customer management, dashboard, ecosystem modules with live API integration
- * Spring Boot: account-opening endpoint, Finacle adapter, scheduled follow-up status job

Inuka Pulse - M&E Intelligence Platform (Expo React Native)

- * Built Case Manager mobile app: KPI dashboard, at-risk beneficiary caseload, dropout risk scoring

INDEPENDENT PROJECTS

Equipment Failure Risk Monitor (Week 10)

Streamlit + XGBoost dashboard scoring 20-machine fleet. ROC-AUC 0.94, Recall 0.83. Reduced maintenance prioritisation from 2 hours to 2 minutes.

Supply Chain Optimisation Dashboard (Week 8)

Multi-source data analysis; optimisation models reduced simulated stockouts by 34%.

ETL Pipeline - Multi-Source Operational Data (Week 4)

Python ETL pipeline across PostgreSQL, CSV, JSON, HTML with Great Expectations validation.

EDUCATION

Certificate - AI & Machine Learning for Operations Power Learn Project (PLP) Academy | 2026

Software Engineering Certificate Moringa School | 2025

Diploma in Information Technology KCA University | 2024